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Chapter 9

Ethical Considerations Surrounding the Use of AI–Interventions in Application of Mental Healthcare

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ABSTRACT

AI's integration into mental healthcare has opened new frontiers in the diagnosis, treatment, and management of mental disorders. This leap brings with it a host of ethical considerations. This chapter provides an analysis of the ethical dimensions of AI interventions in mental healthcare, examining the implications for patient autonomy, privacy, and the integrity of the therapeutic relationship. It discusses the challenges of ensuring informed consent in the context of complex AI systems, the risks of algorithmic bias, its impact on equitable care, and the importance of maintaining confidentiality in the face of expansive data requirements. The chapter

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also explores the need for transparent AI algorithms that can be scrutinized and understood by both practitioners and patients. Furthermore, it addresses the ethical responsibility of AI developers and healthcare providers to prevent harm and ensure the beneficence of AI applications. By highlighting these ethical issues, the chapter aims to foster a responsible approach to the development and deployment of AI interventions.

INTRODUCTION

Envision the future where individuals have immediate access to tailored mental health care, and where artificial intelligence not only identifies and addresses mental health disorders but also anticipates their emergence. This is not a depiction of a distant future, but rather a description of the fast-advancing state of mental health care today. AI treatments have the capacity to fundamentally transform our comprehension, approach, and resolution of mental health issues. Artificial intelligence, sometimes known as AI, is a technology that enables computers and robots to imitate human intellect and problem-solving capabilities. Russel & Norvig (2021) provided a concise definition of AI as an agent that is deliberately created and equipped with the ability to interact with its surroundings, carrying out activities that directly influence the environment. The present period, termed as the “digital revolution,” marks a significant shift in the fourth industrial age. This revolution is characterized by the integration of several technical types, after the mechanical, electrical, and internet eras (Pang et al., 2018; Schwab 2017). Artificial Intelligence (AI) has seen substantial transformations in recent years and is attracting worldwide recognition. Nevertheless, the discipline is expansive and resists simple categorization, since its extent is always evolving owing to swift advancements in this domain.

Artificial intelligence (AI) is revolutionising mental healthcare by offering novel approaches to identify, diagnose, and treat mental health conditions. Machine learning, a branch of artificial intelligence, has the ability to analyse human behaviour patterns, detect indications of mental health issues, identify risk factors, forecast the evolution of illnesses, and customise therapies (Thieme et al., 2020). AI is used in mental health care for several purposes. One use is computational psychiatry, which involves using quantitative methods to evaluate and treat mental disorders (Pandey & Misra, 2023). Another application is incorporating AI into mobile health apps to give mental health help (Graham et al., 2019). Moreover, the potential of artificial intelligence (AI) in mental health care involves using electronic health records, mood rating scales, brain imaging data, innovative monitoring systems like cell phones and video, and social media platforms to predict, categorise, or classify mental health disorders (Milne-Ives et al., 2022). The increasing accessibility of

digital health data creates a foundation for artificial intelligence (AI) to transform healthcare, including the area of mental health care. This sector has been affected by a significant growth in the occurrence of mental health issues, particularly among teenagers (Wies et al., 2021). AI technology may enhance the ability of mental health practitioners to focus on the human aspects of medicine, which can only be accessed via the clinician-patient interaction, while also improving the effectiveness of treatment techniques (Kim et al., 2019; Topol, 2019).

The World Health Organisation (WHO) recommends exercising prudence while implementing AI technology, emphasising the need of thorough review, professional oversight, openness, and active involvement of the public to cultivate confidence in AI. The World Health Organisation (WHO) recommends evaluating clear and indisputable evidence of the advantages before implementing broad usage in healthcare (World Health Organization, 2021). This approach aims to address issues related to safety, individual decision-making, protection of data privacy, adherence to regulatory standards, potential rise in costs, fairness, potential bias in data sources, openness, and accountability. Neglecting to take action in this regard might exacerbate current disparities in society, underscoring the need of using ethical and efficient AI in healthcare (Lee & Yoon, 2021; Rajaram et al., 2023; Sallstrom et al. 2019). It is crucial to guarantee that the advancement and execution of these technologies are carried out properly, taking into account ethical implications and prospective hazards (Obasa, 2023).

The objective of this chapter is to present a comprehensive analysis of artificial intelligence (AI) technologies and their potential uses in mental health care. It will assess the efficacy and constraints of AI-driven interventions in various areas such as diagnosis, therapy, and prevention, while primarily exploring the ethical implications and concerns associated with these interventions. It is essential to thoroughly examine the ethical ramifications and issues related to the use of AI in mental health treatment, including privacy, prejudice, and autonomy. Another primary objective is to identify techniques for incorporating AI technology into current mental health care systems and effectively tackling implementation problems. Finally, we provide guidelines for the responsible advancement and implementation of AI interventions in mental health care, guaranteeing fair and ethical provision of services, as a fundamental objective.

AI TECHNOLOGIES IN MENTAL HEALTH CARE

AI technology has the capacity to transform mental healthcare by offering tools for forecasting, categorization, and customisation of therapy. It has the ability to provide an objective definition of mental diseases, detect them at an earlier stage,

and customise therapies based on individual features. While artificial intelligence (AI) is extensively used in the field of physical health, its implementation in the realm of mental health remains constrained (Ray et al., 2022). Artificial Intelligence is a distinct field of study within computer science that focuses on creating intelligent computers capable of performing tasks related to human cognitive abilities (Beg & Verma, 2023). The system has several elements, such as pre-established regulations, machine learning algorithms, natural language processing, and robots. Machine learning (ML) is the field of research that focuses on algorithms and statistical models designed to improve the performance of computer systems in completing tasks. It may be categorised into two distinct groups: Supervised and Unsupervised. Artificial Neural Networks (ANNs) are machine learning models that are designed to mimic the structure and functions of the human brain. Deep Learning (DL) algorithms have the ability to autonomously acquire knowledge from raw data, without the need for human involvement, hence revealing hidden relationships. Natural Language Processing (NLP) is concerned with the interaction between human language and computer systems, with the goal of enabling computers to understand, interpret, and generate human language. An expert knowledge system employs artificial intelligence to address intricate problems and streamline decision-making procedures (Beg & Verma, 2023).

POTENTIAL APPLICATIONS OF AI IN MENTAL HEALTH CARE

Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have been effectively incorporated into several domains, including the healthcare sector. The efficacy of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) relies on the accessibility of data (Panesar & Panesar, 2021). With the increasing availability of mental health data, we may use AI approaches to improve our comprehension of mental health (Bak et al., 2022).

Recently, deep learning algorithms have shown remarkable outcomes in the diagnosis, prognosis, and treatment of medical conditions in the healthcare sector (Bakator & Radosav, 2018). AI is used in the medical domain using machine learning models to analyse medical data, leading to enhanced health outcomes and patient experiences. AI is used by mental health professionals to accomplish several tasks, including analysing therapy sessions to enhance their effectiveness, improving diagnostic accuracy, monitoring patient well-being, and making required adjustments to treatment plans (Jin et al., 2023).

MENTAL HEALTH DISORDER DIAGNOSIS

Psychiatrists often work under heavy workloads, which might result in a decline in their dependability and accuracy (Ayano et al., 2021). This is evident in a significant prevalence of incorrect diagnoses, mainly due to the fact that the diagnostic procedure substantially depends on individual expertise and subjective evaluations, without the backing of strong biological indicators. The diagnosis of mental health disorders mostly depends on the doctor-patient contact and the assessment of scales. This technique has evident limitations, such as the propensity for patients to refuse acknowledging their symptoms, a deficiency in sensitivity, the existence of subjective biases, and the potential for mistakes. Moreover, there is a notable scarcity of psychiatrists in many regions, especially in developing countries, which impedes the capacity to effectively meet local needs (Yan et al., 2023).

Early detection of mental health issues is crucial for increasing the quality of healthcare and advancing our knowledge of these conditions (Colizzi et al., 2020). However, the task of identifying mental health disorders is rather challenging since the distinction between mental and physical health is sometimes unclear (Costello, 2016). The diagnosis of mental diseases relies mostly on an individual's self-disclosure in response to targeted questions that try to identify certain patterns of social interactions and emotional states (Salamanca-Buentello et al., 2020). In contrast to other chronic illnesses, this procedure does not need laboratory testing or measures. A common challenge in the field of mental health therapy is the lack of mechanistic models for psychiatric diseases. The intricate nature of the brain presents difficulties in developing a conclusive model that clarifies the development of a certain mental condition (Andrew et al., 2023).

The use of machines for the diagnosis of mental diseases has many benefits, such as the preservation of human resources, the improvement of efficiency, the facilitation of large-scale assessments, and the reduction of stigma associated with sickness (Yan et al., 2023). Therefore, a neutral and mechanised method to aid psychiatrists in detecting mental diseases is becoming more and more crucial. Artificial Intelligence (AI) is a potent instrument for identifying medical illnesses, namely in the realm of mental health. Machine learning algorithms such as Boltzmann machine, support vector machine (SVM), and K-Nearest Neighbour (kNN) are used for the purpose of identifying and diagnosing illnesses (Lee et al., 2021; Mahendran et al., 2020). AI subsets, like as computer vision, are extensively used for their strengths in imaging, segmentation, and prediction. Machine learning methods prioritise the capacity of models to perform well on fresh data, which is referred to as generalizability. Computer vision is used to identify, separate, and categorise pictures, including radiological images and metastases.

Artificial intelligence has the ability to distinguish between diagnosis of illnesses that have similar symptoms but need distinct ways of treatment. Utilising data-driven artificial intelligence techniques, such as analysing demographic variables, neurocognitive abilities, and biomarker profiles, may assist in the identification of previously unknown disease subcategories (Shatte et al., 2019). AI techniques can analyse long-term data to identify trends, which is essential for confirming diagnoses by considering the progression of mental illnesses over time.

After examining the wide-ranging effects of AI on diagnosing mental health illnesses, it is important to now concentrate on the practical and concrete ways AI may be used in this particular area. Considerable focus has been given to the development and evaluation of artificial intelligence (AI) models that can detect various mental health diseases in current research. A significant majority of these research have focused on Mood Disorders, including Major Depressive Disorder and Anxiety Disorders (Ding et al., 2018; Shin & Bae., 2023; Tanir & Karakas, 2023). It is important to mention that these AI models have shown encouraging outcomes in the field of mental health diagnoses. They have successfully discerned patterns and accurately forecasted mental health disorders with a high degree of precision.

Zhang et al. (2020) created a Multimodal Deep Learning Framework to identify mental diseases, especially bipolar disorder and depression. The system utilises audio-visual and textual data obtained from interviews. It combines the Multimodal Deep Denoising Autoencoder (multi-DDAE) and Paragraph Vector (PV) models to classify the data using a Multitask Deep Neural Network (DNN). It exhibited similar performance to the approaches that already exist. The research emphasises the possibility of uncovering significant biological indicators for mental problems. Numerous studies have examined the diagnosis of various disorders, including Schizophrenia (Appaji et al., 2019; Karthik & Sudha, 2020; Ko & Yang, 2022; Prabhakar et al., 2020; Tikka et al., 2020), ADHD (Cicek & Akan, 2021; Ding et al., 2018; Kuang et al., 2024; Loh et al., 2022; Maniruzzama et al., 2022), Alzheimer's disease (Vinutha et al., 2020), dementia (Bansal et al., 2018; Bhagyashree et al., 2018), Autism (Hilal et al., 2022), Temporal Lobe Epilepsy (Rajagopalan et al., 2018), and even the detection of post-partum depressive symptoms (Andersson et al., 2021) and burnout (Gupta et al., 2021).

Although AI has the potential to greatly assist in diagnostic procedures, it is important to recognise its current limitations, areas that need more investigation, and the ethical considerations involved. For example, the identification of mental diseases like depression is very subjective, including intricate symptoms, individual variations, and significant socio-cultural connections (Vares et al., 2015). Hence, further effort is necessary to narrow the divide between artificial intelligence (AI) in mental health research and its use in clinical treatment.

Furthermore, the present scope of AI technologies in healthcare decision-making and patient care is restricted due to low practical application and clinical validity. Computer scientists devote significant effort to creating computationally efficient and robust models that are customised for certain databases. The accuracy of diagnosing mental health disorders is primarily limited by the quality of the samples and the accuracy of their annotations in the datasets. This, in turn, affects the effectiveness of AI tools, with overfitting being a common problem when using small and restricted datasets. The representativeness of the sample often decreases when factors such as group affiliations, language origins, and cultural settings are taken into account. Concerning the sample size, each database has a limited number of data samples, which is a restriction imposed by ethical concerns that are crucial for dataset release.

The presence of heterogeneity makes it more difficult to demonstrate validity across various datasets. The particular aspects of imaging data, such as how the data is obtained, the settings used during scanning, and the procedures used for processing, hinder the ability to apply these findings to other datasets. This makes it difficult to create meaningful comparisons based on the results. The wide-ranging usefulness of machine learning models is also constrained by their assessment only based on in-sample data, without considering out-of-sample data.

The predictive efficacy of these research is also limited by the input elements, such as clinical data and demographics. Considering that it is impossible for any research to include all factors, it is important to consider the clinical effectiveness of the characteristics utilised to create the models. The algorithm's results may have limited application to certain groups or scenarios. According to Yan et al. (2023), the use of artificial datasets with limited sample sizes in AI learning leads to a lack of ecological validity. As a result, the concept of "AI diagnosis" becomes a logical contradiction and a self-fulfilling prophecy. Consequently, the precision of AI diagnosis is significantly diminished when confronted with ecological clinical data.

In addition, there are certain challenges that arise from the existing diagnostic criteria and traditional diagnostic techniques, including the difficulties associated with identifying depressive illnesses because of their subjective character and the variation in symptoms across different people. The DSM-5 criteria, which include primary symptoms including a diminished mood and decreased interest, may not consistently apply to every patient, leading to the emergence of diverse depression profiles (Tolentino & Schmidt, 2018). Moreover, the presence of comorbidities and the intermittent presentation of symptoms provide additional levels of intricacy to the diagnosis, presenting substantial obstacles for AI in precisely detecting and diagnosing depressive illnesses using these criteria. In order to support the claim, the accuracy of the diagnosis relies on the patient's capacity to effectively convey their symptoms and the clinician's level of expertise, highlighting the need of thorough information and clinical training for precise diagnosis. Biological markers are taken

into account, although they are not absolute indications, highlighting the intricate nature of the diagnostic procedure.

The increase in utilisation of web servers poses a threat to the confidentiality of patient information. Online databases and associated storage devices are vulnerable to unauthorised access and cyber-attacks. In India, the Ministry of Health and Family Welfare has proposed the Healthcare Security Act, which includes provisions for penalising data breaches and promoting the use of electronic records. However, there is a clear lack of regulation regarding the assistance given to mental health practitioners in offering AI services. There are currently no legislative regulations in place to impose responsibility on software developers for technical faults (Ray et al., 2022).

Moreover, the management of mental health relies on interpersonal aptitudes such as establishing a rapport and cultivating associations with patients, while also monitoring their emotional and behavioural tendencies. The absence of crucial components like human compassion and empathy is a substantial barrier in the therapy procedure. Further research into AI applications is crucial in order to mitigate unforeseen tendencies. Prior to incorporating technology into medical practice, a thorough risk assessment and administrative oversight are necessary to ensure its capacity to handle uncommon situations.

AI FOR THERAPY

During the 2000s, the use of Information and Communication Technologies (ICTs) in Clinical Psychology became more common. Virtual Reality (VR) and Telepsychology have become integral to the area, with studies specifically examining the efficacy of VR in cognitive behavioural treatment (CBT) for different mental diseases and health issues. The capacity of virtual reality (VR) to replicate reality has substantial significance in the industry. However, it also gives rise to ethical quandaries, such as the possible consequences of blurring the distinction between actual and virtual worlds, as well as worries around cybersickness. Telepsychology has been used to augment the administration of cognitive behavioural therapy (CBT), with internet-based programmes emerging as crucial instruments in the healthcare system. Nevertheless, there were apprehensions over self-help methods, including self-diagnosis and patients' ability to get all essential self-help information (Botella et al., 2009).

AI has become the dominant force in the 21st century, sparking debates and discussions about its effectiveness and importance in various fields, including psychotherapy. Its use in treatments has garnered attention and raised questions at both personal and societal levels. This has not only led to extensive debate but

has also opened up new possibilities for mechanising psychotherapy. Healthcare delivery has expanded beyond the limitations of the waiting room and has really become mobile (Panesar & Panesar, 2021). Numerous research papers have explored the capacity of AI to either aid or function as an active agent in the provision of psychotherapy to patients. Research suggests that the COVID-19 pandemic caused widespread emotional distress and increased the likelihood of mental disorders. These disorders may be directly related to the COVID-19 disease or indirectly caused by strict public health measures that limited personal freedoms and resulted in financial difficulties. Furthermore, the pandemic had a disproportionate effect on those with serious mental health disorders. The impact of this phenomenon was particularly disruptive for young people, with a considerable proportion of them reporting a rise in mental health issues as a direct result of the lockdown measures (Anand et al., 2021).

An effective approach to mitigate the medical and economical consequences of mental illnesses is to use the capabilities of digital health technology to provide supportive, preventative, and therapeutic interventions for those requiring assistance. As a result, digital mental health has become a rapidly growing field of study in the domains of digital health and scientific research. The primary customer or patient population for digital mental health services is largely composed of young people, who are leading the way in adopting all elements of digital technology, including digital health. Mounting research suggests that digital mental health interventions have the capacity to improve mental health conditions, such as depression, in various populations of patients. The beneficial capability of digital mental health technology for continuous and accessible provision of care is especially relevant to mild-to-moderate cases that are not being evaluated and treated, particularly in areas where mental health care resources are under significant pressure due to severe mental illness cases (Wies et al., 2021).

The mental health industry is undergoing a significant change with the introduction of Artificial Intelligence (AI)-enabled treatment, also known as AI-supported therapy. This emerging discipline utilises cutting-edge technology like machine learning and natural language processing to provide therapeutic treatments. AI systems can analyse patient data, detect trends, and suggest therapy plans, so improving the effectiveness of human therapists. An important benefit of AI treatment is its capacity to give 24/7 support, providing instant assistance in emergency situations. This function is especially advantageous in areas with a limited number of mental health specialists, so helping to reduce the disparity in access.

Furthermore, AI is crucial in automating the monitoring of clinical indicators, which may include a wide variety of activities such as online interactions and gamified therapies. The advancement of this discipline requires a cooperative endeavour between mental health experts and technological specialists. This collaboration

is essential for facilitating more progress in this field. Chatbots are software programmes that use Artificial Intelligence (AI) to imitate human-like interactions. They are often used on platforms that offer messaging or voice chats. Their use in the field of mental health has been steadily growing, embracing responsibilities like as providing companionship, support, and therapeutic treatments.

Chatbots provide several advantages as compared to traditional in-person or digital interactions. These factors encompass improved accessibility, which ensures that therapeutic support is accessible to a broader range of people; affordability, which reduces the financial obstacles to mental health care; anonymity, which creates a secure environment for individuals to freely express their emotions without concern for criticism; and scalability, which enables the provision of support to multiple individuals simultaneously. This novel approach to mental health treatment represents a hopeful change in the delivery of therapeutic services. Significantly, several chatbots have been created to tackle both the scarcity of mental health experts and the increasing need for digital alternatives to conventional face-to-face treatment.

Woebot, an automated conversational agent, has been created to provide Cognitive Behavioural Therapy (CBT). It has been deemed trustworthy by multiple researchers, particularly in its ability to alleviate depressive symptoms (Balcombe, 2023; Fitzpatrick et al., 2017; Ly et al., 2017; Nicol et al., 2022; Pham et al., 2022). Additionally, it has been proven effective for treating Substance Use Disorders (Prochaska et al., 2021a; Prochaska et al., 2021b). Furthermore, Wysa, Tess, XiaoNan, and TEO, among other chatbots (Leo et al., 2022; Fulmer et al., 2018; Liu et al., 2022; Danieli et al., 2022), have demonstrated efficacy in treating patients with mood-related disorders and Substance Use disorder.

These chatbots have been praised for their convenience, affordability, and easy accessibility, making them viable alternatives to traditional psychotherapy. However, the inherent lack of transparency in the underlying algorithms poses a barrier in predicting the development of these discussions. Existing behavioural studies provide only a limited comprehension of the possible safety issues that might emerge from the rapid and widespread use of this technology. Furthermore, chatbots do not possess the extensive talents that are inherent in a psychiatrist or therapist that has had professional training. Their ability to use individual patient data to assist with the patient's cognitive tasks is limited. In addition, they may lack the nuanced emotional comprehension and compassionate response that is typical of their human counterparts. Currently, there are no direct applications in the therapeutic setting, and there is no established criterion for measuring technical progress. An important concern is around legal liability, prompting the question of who would assume responsibility in the event that the artificial intelligence entity makes a wrong diagnosis or misinterprets a patient's suffering. Although there have been significant technology breakthroughs in this area, mental health practitioners have been slow

to accept new technologies. Psychiatrists who prioritise personal connections with patients may have a slower rate of acceptance, suggesting a gradual spread of new ideas throughout the mental health field. Furthermore, the exponential increase in the use of mental health apps adds complexity to the process of assessing risks, which might result in evaluations being conducted after harm has already occurred (Coghlan et al., 2023).

RISK PREDICTION

Risk assessment is often regarded as an essential element in the management of self-inflicted injury and aggressive conduct. Within the field of psychiatry, the objective of risk assessment is to mitigate the negative consequences linked to mental diseases. The technique involves categorising patients into various risk groups according to the likelihood of self-harm or aggressive conduct, with the aim of providing guidance for treatment and management choices (Ryan et al., 2010). The use of risk assessment is increasingly promoted to reduce occurrences of self-harm and violence. On the other hand, the efficiency of risk assessment has been a topic of ongoing and thorough debate. Critics contend that there is no empirical evidence to substantiate the claim that any kind of risk assessment may effectively alleviate the adverse consequences associated with mental health issues. Recently, precision psychiatry has emerged as a promising approach to accurately estimate individualised risks for various clinical outcomes.

Latest studies indicate that specific AI systems could aid in assessing the probability of psychosis onset (Oliver et al., 2021). Initiatives have been undertaken to classify recognized risk elements, enabling medical professionals to sort individuals into categories of lower and higher risk. These implications have significant consequences for treatment choices, such as determining whether to admit a patient to a hospital, either by their own choice or against their will. A considerable proportion of artificial intelligence (AI) endeavours in the field of mental healthcare have been focused on tackling the issue of suicide. Machine Learning (ML) systems have the ability to detect risk factors for suicide by analysing complicated relationships between variables, rather than just looking at them individually. ML approaches used to well-structured datasets have shown connections between suicide risk and clinical characteristics such as ongoing depression, past psychiatric diagnosis, drug use, and treatment history (Passos et al., 2016). Further analysis has shown the influence of environmental variables (Bonner & Rich, 1990; Fernandez-Arteaga et al., 2016).

Researchers have used multi-level modelling of Internet search trends to identify relevant risk variables (Song et al., 2016). Google's data analytics have, in some cases, yielded a more precise assessment of suicide risk compared to conventional

self-report scales (Ma-Kellams et al., 2016). Researchers from Vanderbilt University have created an algorithm that can predict the likelihood of suicide by analysing demographic and personal health data obtained from Electronic Medical Records (EMRs). The accuracy of this model in predicting suicide risk was 84-92% within a week and 80-86% within two years, according to Walsh et al. (2017). Comparable results were obtained when Machine Learning was used to forecast the likelihood of teenage suicidality, by amalgamating more than 600 risk indicators retrieved from Electronic Medical Records (EMRs). The precision of the predictions grew as the likelihood of suicide attempts became more imminent (Walsh et al., 2018). Concurrently, scientists at the University of Vermont developed a social media algorithm to identify indicators of depression. This was accomplished by doing a linguistic content analysis of user accounts, with a special emphasis on word count, language use, voice patterns, and activity level (Holley, 2017). BiAffect, a technology developed by the University of Illinois Chicago, employs an iPhone app to scrutinize typing patterns and emotional conditions in individuals suffering from bipolar disorder or major depressive disorder. The Mobile Typing Kinematics method can predict mental health problems and improve patient autonomy, accessibility of treatment, and cost-effectiveness. Digital phenotyping, often known as personal sensing, uses data from personal digital gadgets, particularly smartphones, to anticipate psychological outcomes and mental health disorders. This data, when processed through machine learning techniques, can aid in the selection of effective strategies. Clinical studies are using digital phenotyping to identify first alterations in depression, schizophrenia, and PTSD. Signals obtained from phones or wearables have the potential to enhance or substitute traditional clinical evaluations (Järvelä et al., 2022). Digital phenotyping is also used for monitoring patients with severe mental diseases, detecting a “early warning signal” that indicates the onset of a relapse, therefore facilitating care management.

The integration of Artificial Intelligence (AI) into prediction models and clinical procedures requires addressing privacy concerns, particularly those related to the safe collection, storage, and transfer of Personal Health Information (PHI). The use of internet-based and social media platforms presents inherent security risks that are presently not regulated by existing privacy legislation. The inherent synchronisation features may unintentionally result in the spread of user data across several apps or to third-party entities. The lack of a clear differentiation between private and public material creates challenges in establishing data access limits, especially when anonymized Protected Health Information (PHI) is linked to identifying information. Twitter and Google have faced criticism for violating privacy standards, leading to privacy infringements that may cultivate distrust in the mental healthcare system and discourage patients from seeking assistance. In order to reduce these dangers,

it is crucial to expand the range of protective laws for Protected Health Information (PHI) to include possible hazards related to Artificial Intelligence (AI).

The accuracy of Artificial Intelligence (AI) pattern recognition systems in detecting suicidal thoughts gives rise to worries over the possibility of misdiagnosis. This highlights the significance of algorithmic training in correcting mistakes and improving the confidence of the model, while also ensuring optimum processing speed. Moreover, prior to being made available for public use, these algorithms must undergo validation in many situations, since their prediction accuracy may vary across distinct environments. Therefore, it is crucial to train algorithms in order to provide accurate and reliable classification of people (Mulder, 2016).

The significance of ensuring patient safety is of utmost relevance, especially when considering the use of AI technologies. It is crucial to ensure that these programmes can effectively handle users who indicate suicidal inclinations, without causing emotional distress or contributing to thoughts of suicide. The unpredictable and often inappropriate replies from smartphone voice assistants when asked about mental health might have a negative impact on safety concerns and discourage users from seeking help or discussing personal problems. The issue of legal responsibility is a major worry in healthcare settings when incorporating new AI technology (Khosravi et al., 2024). It is crucial to develop standards for evaluating patient eligibility, monitoring outcomes, handling high-risk situations, and controlling costs. Response plans must be developed to address high-risk scenarios, erroneous alerts, and conflicting risk assessments. Moreover, it is essential to establish clear criteria regarding culpability in cases of suicide.

ETHICAL IMPLICATIONS AND CONCERNS

As previously said, it is clear that AI has the capacity to greatly impact the field of mental health treatment. However, it is important to acknowledge that despite the increasing role of AI in mental health care, addressing various issues such as early detection, diagnosis, therapeutic interventions, cost-effective support, personalised treatment, automation of tasks through digitalized healthcare data, and enhancing our understanding of complex disorders, there are significant shortcomings. One of these is the overly accelerated promotion of new AI models that have not yet been evaluated for their viability in real-world settings. The ethical issues in data science may be classified into three primary domains: the ethics pertaining to data management, the ethics linked to artificial intelligence, and the ethics surrounding the methodology used (Egger et al., 2022). The development of machine learning technologies is giving rise to moral and ethical concerns. The administration in this sector is growing at a parallel rate to the advancement of the industry. In the field

of AI, there are many instances where there is a lack of established precedents, regulations, or laws (Gordon, 2021). Hence, it is essential for both developers and practitioners to carefully consider the moral and ethical implications of creating intelligent systems.

Bias

One of the most prominent ethical problems associated with the use of AI is **bias**, which may manifest in several ways:

- A. Machine Learning bias, also known as algorithmic bias or AI bias, occurs when an algorithm consistently generates biased results as a result of faulty assumptions made during the machine learning process. This bias may originate from several sources, including the partiality of the designer shown in the data used to train the model. These biases often arise as novel concerns with both nuanced and evident ramifications. Every machine learning algorithm relies on a statistical bias to create predictions about unfamiliar data. Machine bias, however, signifies a bias shown by the data creators. Due to the superior processing speed and capacity of AI compared to humans, it is not always dependable for impartial and unbiased outcomes.
- B. Data bias occurs when the data obtained does not correctly represent the population or event being studied. It might also suggest that the data is missing the critical characteristics to accurately predict the event we are trying to foretell.
- C. Artificial intelligence systems are created by people, who inevitably have biases and make judgements. Hence, it seems likely that as long as humans continue to be involved in decision-making processes, the existence of bias is unavoidable.
- D. AI systems that replicate human cognition have been shown to amplify prejudices, leading to concerns regarding the ethical consequences of AI use. Early cognitive systems constructed utilising demographic data had significant bias about gender, ethnicity, financial level, and other characteristics.

Although it is difficult to totally eliminate prejudice, it is important to recognize that biases might arise in AI systems. Hence, AI developers must be aware about this potential danger and make efforts to reduce biases at every stage of model development. This involves considering possible biases when choosing machine learning technology and methods for training algorithms and programming datasets. Furthermore, augmenting the variety of data might assist in mitigating biases in risk models that rely on machine learning. Simulated model co-creation is a method that promotes responsible modelling by considering the effects of simulations and developing user narratives. In order to address bias in AI research, it is recommended

to use approaches such as oversampling, under sampling, and module penalization score (Chauvin et al., 2016; Golbin et al., 2020; Landau et al., 2022; Luxton, 2020; Panesar & Panesar, 2021).

Fairness

The ethical dimension of fairness in AI systems is closely linked to prejudice. This entails the imperative task of ensuring that artificial intelligence does not exhibit favouritism towards any particular group or inflict damage onto certain persons or groups as a result of biased data or algorithms. Ensuring equality and justice are fundamental ideals that must be upheld in the design, development, and implementation of AI. Ensuring equitable and fair treatment in government regulations and business practices is crucial when introducing new technology and services to neglected populations. The covert algorithms, data analysis systems, and profiling operations of the Big-Data engine should be prohibited from engaging in discriminatory practices. Moreover, the absence of technology awareness and education might limit access to conversational agents such as chatbots. When faced with these situations, it is advisable to explore in-person or community-based programmes as viable options to tackle this disparity in access. The COVID-19 pandemic has highlighted the significance of health equality, given the growing digitization of public health. It is important to ensure that the integration of digital innovations in public health does not unintentionally worsen health inequalities related to socioeconomic level (Al-Hwsali, 2022; Beauchamp, 2008; Beauchamp & Childress, 1994; Karimian et al., 2022; Kim et al., 2021).

Privacy and Security

Expanding upon the discourse around bias, privacy and data security emerge as notable ethical considerations in the field of AI. Given that AI systems often need substantial quantities of data for training and functioning, safeguarding the privacy and security of this data is of utmost importance. This encompasses the protection of personal information and the prevention of unauthorised access or data breaches. It is essential to include security and privacy protocols into the AI development process to guarantee the safeguarding of all AI systems, data, and health-related interactions. Patient data is the fundamental basis for the advancement of artificial intelligence in the field of healthcare. When scholars explore the ethical implications of Artificial Intelligence (AI) systems, they focus heavily on the privacy, confidentiality, and security of user data and information collected via digital mental health apps. Scholars are primarily concerned about the possible consequences that may arise if secret material is shared or made available to other

parties. Researchers have contended that insufficiencies in data security while sharing information might have negative consequences that extend to many areas of life, such as work settings, educational establishments, and personal connections with friends, family, or lovers. Studies have been carried out on wearable fitness devices that collect and analyse user data, with the potential to uncover risk factors for many public health problems. These investigations emphasise the possibility of unauthorised access to confidential information, prompting concerns about the proper management of these devices to protect patient privacy and avoid breaches that might have ethical and legal consequences. Users should be informed about any privacy limitations imposed by the administrator. Moreover, it is crucial to acknowledge that research using AI models created using open-source or democratised methods might be compromised by malevolent individuals. It is crucial to develop strong and clear norms, in collaboration with the public health sector, to protect patient confidentiality (European Commission, 2019; Fiske et al., 2019; Morley et al., 2020; Reddy et al., 2020).

The World Health Organisation has established clear rules on this matter. Healthcare personnel have a duty to protect sensitive information, regardless of the manner in which it is kept or received. They should be aware that their professional encounters might establish the limits and regulations of secrecy. In their 2022 publication, Bak et al. examined the ethical quandary of reconciling health data privacy and accessibility within the realm of AI advancements in the healthcare sector. The authors emphasised the inherent trade-offs between safeguarding individual data and the need of using extensive datasets to train AI systems, hence enhancing healthcare outcomes. The authors contend that allocating significant resources to AI development while simultaneously imposing stringent privacy controls that limit data access is both morally wrong and inefficient. Their objective is to support a just and comprehensive approach in establishing national data governance regulations that are in line with public values and goals, so assuring the responsible development of healthcare AI.

Kretzschmar et al. (2019) examined the ethical ramifications of using chatbots to provide mental health assistance to adolescents. The Oxford Neuroscience, Ethics and Society Young People's Advisory Group, including individuals aged 14 to 18, deliberated about the advantages and disadvantages of chatbots. In the context of data gathering and privacy, the authors suggest that the following principles should be followed:

1. It is essential to ensure the protection and confidentiality of any personal information that is acquired.
2. If there is a need to disclose the content of talks, it should be anonymized to guarantee anonymity.

3. Users must be provided with unambiguous and thorough information about privacy measures and any corresponding restrictions.
4. Users should have the option to seek reminders about these privacy arrangements and constraints at any time throughout their contact. This guarantees a constant state of consciousness and comprehension about their rights concerning the privacy of their data.

The “Black Box” Issue

Moreover, the issue of transparency emerges. AI systems, especially those using deep learning, are often known as “black boxes” because of their intricate and obscure decision-making mechanisms. The absence of transparency may result in distrust and presents difficulties in ensuring accountability. Understanding the logic or prediction principles of an AI system may be difficult. The issue of the “black box” is considered a major obstacle to ensuring transparency in the healthcare industry. The absence of transparency may hinder the ability of healthcare professionals and patients to engage in meaningful discussions on the advantages and disadvantages of following a choice made with the assistance of AI. AI-driven diagnostic tools may analyse medical images, such as X-rays, MRIs, and CT scans, to identify patterns. Nevertheless, the lack of openness in the algorithm's decision-making process hinders the verification of the tool's accuracy and the comprehension of its diagnostic technique. Enhanced transparency in the decision-making process promotes better governance by offering a more comprehensible insight into the utilisation and implementation of models. The assignment of responsibility in AI is complex, mainly because of the absence of transparency (Anderson & Anderson, 2019; Braun et al., 2021; Buruk et al., 2020; Liyanage et al., 2019).

Consequently, people may experience a loss of the capacity to oversee decision-making processes and may find it difficult to understand the rationale behind less-than-ideal decisions. Hence, it is important to comprehend the decision-making methods used by this technology.

Responsibility and Accountability

Ultimately, the ethical concern of utmost importance is the responsibility and accountability for judgements made by AI systems. Addressing the subject of accountability for mistakes or damage caused by an AI system is a challenging and intricate matter. The ethical dilemma with accountability in AI systems mostly centres on the lack of clearly defined criteria for accountability or corresponding ethical obligations to protect users who are using mental health services. Unlike traditional therapy, which is obligated by ethical responsibilities to safeguard the

client's welfare, AI therapy apps sometimes lack proper professional oversight. This prompts inquiries about the organisation accountable for ensuring the user's security, confidentiality, and general welfare. Accountability is essential in the field of public health technology, namely in the creation, advancement, and use of AI-driven solutions including health promotion, illness prevention, treatment methods, and disease diagnostics. It is crucial to carefully examine these judgements in order to determine their suitability for the general public. The obligation is significant and mostly lies with the man in charge of these activities. Therefore, it is essential for all healthcare workers to improve their expertise in managing AI technologies. It is crucial to avoid mistakes that have the potential to greatly affect the health of the community (Horvitz & Mulligan, 2015; Matthias, 2004; Nebeker et al., 2019; Omrani et al., 2022; World Health Organization, 2021).

Ultimately, while AI has the promise of transforming several facets of human existence, it is imperative to address these ethical dilemmas in order to guarantee its conscientious and advantageous use. As we strive to make advancements and expand the capabilities of AI, it is imperative that we consistently evaluate and tackle the ethical concerns associated with it.

RECOMMENDATIONS FOR RESPONSIBLE DEVELOPMENT AND DEPLOYMENT

In order to ensure the responsible development and implementation of AI in mental health care, it is crucial to adopt a comprehensive strategy that places emphasis on ethical principles, fair provision of services, and cooperative resolution of issues. The ethical principles for AI developers and practitioners should emphasize the significance of openness, informed consent, and privacy. It is crucial to ensure that AI systems are created with a deep regard for patient autonomy and confidentiality. Furthermore, it is essential to establish policies that ensure fair and impartial delivery of AI services in order to address and reduce inequalities in access to mental health treatment. These policies should focus on bridging the gap in access to digital resources and aim to develop AI tools that are culturally sensitive and available to a wide range of communities.

Finally, the use of collaborative methods is crucial for overcoming obstacles in the implementation process. This entails cultivating collaborations among technologists, physicians, patients, and politicians to collaboratively develop solutions that are both technically viable, therapeutically applicable, and socially agreeable. These coordinated endeavors will not only increase the reliability of AI applications but also guarantee their conformity with the main objective of enhancing mental health outcomes.

CONCLUSION

At the junction of technical advancement and mental health care, the emergence of Artificial Intelligence (AI) presents unparalleled prospects for improving the quality and availability of therapy. Nevertheless, this auspicious future is not without of its ethical dilemmas. Incorporating AI into mental health treatments requires a careful and ethical strategy to prioritise the well-being of patients. The ethical implications of artificial intelligence in the field of mental health are many. The preservation of privacy and secrecy is of paramount importance, given the extensive data necessary to fuel AI systems, which has the potential to jeopardise sensitive personal information. Implementing strong data protection measures is essential for sustaining the trust that is the basis of any therapeutic engagement.

Obtaining informed permission is an essential aspect of ethical conduct. Patients must be informed on the specific ways in which artificial intelligence (AI) will be used in their healthcare, including both the possible advantages and disadvantages. The intricate nature of AI systems presents a difficulty in achieving transparency, but, it is a difficulty that necessitates explicit communication and education. AI systems that exhibit bias and lack justice might result in unequal treatment, particularly impacting marginalised populations. Developing and training AI systems on various datasets is crucial in order to reduce biases and ensure fair treatment results. The opaque nature of several AI systems gives rise to concerns over transparency and comprehensibility. Healthcare practitioners must possess the capacity to comprehend and communicate the rationale behind AI-generated suggestions in order to uphold responsibility and maintain patient confidence.

The use of artificial intelligence into mental health treatment also has an effect on the therapeutic interaction. Although AI has the potential to enhance the skills of healthcare professionals, it is crucial to maintain the fundamental human aspect that is vital to the process of healing and recuperation. The growth of AI technology has outpaced the establishment of effective oversight and regulation, resulting in a governance vacuum that has to be resolved. It is essential to establish precise norms and standards for artificial intelligence (AI) in mental health treatment in order to protect the interests of patients. The process of attributing responsibility and accountability for judgements made by artificial intelligence, especially in cases when negative results occur, is intricate. It is crucial to establish a distinct and unambiguous assignment of responsibility between AI developers, healthcare providers, and the technology itself.

The use of AI in mental health treatment has the capacity to revolutionise the field, but it also poses the danger of devaluing the expertise of professionals or compromising the standard of care if employed as a replacement for human decision-making. It is crucial that AI be used as a tool to enhance, rather than replace, professional

skills. Ultimately, cultural sensitivity is of utmost importance in the field of mental health treatment. AI systems should be built to include cultural diversity in order to guarantee that the therapy provided is courteous and suitable for all patients. Essentially, ensuring the ethical use of AI in mental health treatment involves carefully managing the utilisation of technology progress while maintaining the moral obligations of the medical field. As we plan for the future, it is our shared duty to have ongoing discussions and thoughtful thinking, making sure that AI is used in a way that supports mental health in a sincere, caring, and respectful manner for everyone.

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